Supporting Information

Interpretable Machine Learning for Predicting GHS Chemical Hazard Classifications: A Multi-Label Approach Using 243,323 Compounds and Scaffold-Based Validation

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The tables below are also provided as an Excel workbook, *publication_supplementary_tables.xlsx*, in which every table appears in full. The underlying dataset, descriptors and trained models are archived at https://doi.org/10.5281/zenodo.21876611 and the analysis code at https://doi.org/10.5281/zenodo.21903565.

Table S0

GHS label schema: column names, pictogram codes, their authoritative United Nations meanings, and the correspondence with the names used in the original study design.

column name	pictogram code	authoritative UN PubChem meaning	name in original proposal	was renamed
GHS01_Explosive	GHS01	Explosive (exploding bomb)	GHS01_Explosive	no
GHS02_Flammable	GHS02	Flammable (flame)	GHS02_Flammable	no
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	GHS03_Oxidising	no
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	GHS04_CompressedGas	no
GHS05_Corrosive	GHS05	Corrosive (corrosion)	GHS05_Corrosive	no
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and cross...	GHS06_AcuteToxicity	no
GHS07_Irritant	GHS07	Irritant / harmful (exclamation...	GHS07_HealthHazard	YES
GHS08_HealthHazard	GHS08	Serious health hazard (health h...	GHS08_Environmental	YES
GHS09_Environmental	GHS09	Environmental hazard (environment)	GHS09_Irritant	YES

9 rows. Source sheet: S0_label_schema.

Table S1

Full dataset statistics after cleaning: compound counts, percentages and imbalance ratios per hazard class.

GHS Column	Pictogram Code	Actual Meaning	N Positive	N Negative	Percent of Dataset	Imbalance Ratio NegPerPos	Name in original proposal	Note
GHS01_Explosive	GHS01	Explosive (exploding bomb)	184	243139	0.076	1321.41	GHS01_Explosive	
GHS02_Flammable	GHS02	Flammable (flame)	10632	232691	4.37	21.89	GHS02_Flammable	
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	419	242904	0.172	579.72	GHS03_Oxidising	
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	285	243038	0.117	852.76	GHS04_CompressedGas	
GHS05_Corrosive	GHS05	Corrosive (corrosion)	43336	199987	17.81	4.61	GHS05_Corrosive	
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and cross...	15201	228122	6.247	15.01	GHS06_AcuteToxicity	
GHS07_Irritant	GHS07	Irritant / harmful (exclamation...	220020	23303	90.423	0.11	GHS07_HealthHazard	Renamed from the original study...
GHS08_HealthHazard	GHS08	Serious health hazard (health h...	13606	229717	5.592	16.88	GHS08_Environmental	Renamed from the original study...
GHS09_Environmental	GHS09	Environmental hazard (environment)	17735	225588	7.289	12.72	GHS09_Irritant	Renamed from the original study...

9 rows. Source sheet: S1_dataset_statistics.

Table S2

Complete hyperparameter search results for all three algorithms.

Algorithm	GHS Class	Best CV Score	N Iterations	CV Folds	Best Parameters
RandomForest	all (multi-output)	0.8246	5	3	{"estimator__n_estimators": 500...
XGBoost	GHS01_Explosive	0.8917	5	3	{"subsample": 0.9, "reg_lambda"...
XGBoost	GHS02_Flammable	0.9625	5	3	{"subsample": 0.6, "reg_lambda"...
XGBoost	GHS03_Oxidising	0.9879	5	3	{"subsample": 0.6, "reg_lambda"...
XGBoost	GHS04_CompressedGas	0.9972	5	3	{"subsample": 0.6, "reg_lambda"...
XGBoost	GHS05_Corrosive	0.8565	5	3	{"subsample": 0.6, "reg_lambda"...
XGBoost	GHS06_AcuteToxicity	0.7775	5	3	{"subsample": 0.9, "reg_lambda"...
XGBoost	GHS07_Irritant	0.7775	5	3	{"subsample": 0.6, "reg_lambda"...
XGBoost	GHS08_HealthHazard	0.8831	5	3	{"subsample": 0.6, "reg_lambda"...
XGBoost	GHS09_Environmental	0.851	5	3	{"subsample": 0.6, "reg_lambda"...
SVM	all (multi-output)	0.745	30	3	{"estimator__svm__kernel": "rbf...

11 rows. Source sheet: S2_hyperparameter_search.

Table S3

Full performance metrics: every model, class, metric and decision threshold, with bootstrap confidence intervals.

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
RandomForest	GHS01	default_0.5	0.5	18	0.9722	0.9518	0.9889	0.1184	0.0576	0.1277	0.0304	0.5556
RandomForest	GHS01	calibrated_F1	0.879101	18	0.9722	0.9518	0.9889	0.1184	0.1379	0.1416	0.1818	0.1111
RandomForest	GHS01	safety_recall90	0.152825	18	0.9722	0.9518	0.9889	0.1184	0.0101	0.0659	0.0051	1.0
RandomForest	GHS02	default_0.5	0.5	1063	0.952	0.9457	0.9581	0.6242	0.3864	0.433	0.2456	0.9059
RandomForest	GHS02	calibrated_F1	0.797954	1063	0.952	0.9457	0.9581	0.6242	0.589	0.5702	0.5882	0.5898
RandomForest	GHS02	safety_recall90	0.497792	1063	0.952	0.9457	0.9581	0.6242	0.385	0.4321	0.2443	0.9078
RandomForest	GHS03	default_0.5	0.5	41	0.9321	0.8751	0.978	0.4208	0.1187	0.2038	0.0652	0.6585
RandomForest	GHS03	calibrated_F1	0.93163	41	0.9321	0.8751	0.978	0.4208	0.42	0.4258	0.3559	0.5122
RandomForest	GHS03	safety_recall90	0.205989	41	0.9321	0.8751	0.978	0.4208	0.0393	0.1229	0.0201	0.8293
RandomForest	GHS04	default_0.5	0.5	28	0.9726	0.9102	0.9994	0.5792	0.1289	0.256	0.0691	0.9643
RandomForest	GHS04	calibrated_F1	0.96652	28	0.9726	0.9102	0.9994	0.5792	0.4889	0.5038	0.6471	0.3929
RandomForest	GHS04	safety_recall90	0.851522	28	0.9726	0.9102	0.9994	0.5792	0.4107	0.4733	0.2738	0.8214
RandomForest	GHS05	default_0.5	0.5	4333	0.8448	0.8378	0.8513	0.6571	0.5482	0.4422	0.44	0.7272
RandomForest	GHS05	calibrated_F1	0.561174	4333	0.8448	0.8378	0.8513	0.6571	0.5901	0.4961	0.5568	0.6275
RandomForest	GHS05	safety_recall90	0.351714	4333	0.8448	0.8378	0.8513	0.6571	0.4147	0.2919	0.2685	0.91
RandomForest	GHS06	default_0.5	0.5	1520	0.802	0.7905	0.813	0.2487	0.269	0.2507	0.1687	0.6638
RandomForest	GHS06	calibrated_F1	0.601884	1520	0.802	0.7905	0.813	0.2487	0.3162	0.2669	0.2778	0.3671
RandomForest	GHS06	safety_recall90	0.369084	1520	0.802	0.7905	0.813	0.2487	0.189	0.185	0.106	0.8717
RandomForest	GHS07	default_0.5	0.5	21967	0.8061	0.7978	0.8143	0.9718	0.8518	0.3029	0.9579	0.7668
RandomForest	GHS07	calibrated_F1	0.232387	21967	0.8061	0.7978	0.8143	0.9718	0.9491	0.1748	0.9077	0.9945
RandomForest	GHS07	safety_recall90	0.354999	21967	0.8061	0.7978	0.8143	0.9718	0.9242	0.2804	0.9317	0.9168
RandomForest	GHS08	default_0.5	0.5	1360	0.8645	0.8559	0.8733	0.3053	0.311	0.3109	0.1992	0.7088
RandomForest	GHS08	calibrated_F1	0.686155	1360	0.8645	0.8559	0.8733	0.3053	0.363	0.324	0.3492	0.3779
RandomForest	GHS08	safety_recall90	0.37313	1360	0.8645	0.8559	0.8733	0.3053	0.2326	0.2605	0.1338	0.8875
RandomForest	GHS09	default_0.5	0.5	1773	0.8637	0.8553	0.8721	0.3834	0.3791	0.3644	0.2538	0.749
RandomForest	GHS09	calibrated_F1	0.664859	1773	0.8637	0.8553	0.8721	0.3834	0.4149	0.3662	0.3812	0.4552
RandomForest	GHS09	safety_recall90	0.412591	1773	0.8637	0.8553	0.8721	0.3834	0.3318	0.3325	0.2072	0.8319
XGBoost	GHS01	default_0.5	0.5	18	0.9859	0.9743	0.9953	0.1507	0.0785	0.1713	0.0415	0.7222
XGBoost	GHS01	calibrated_F1	0.912269	18	0.9859	0.9743	0.9953	0.1507	0.1818	0.2355	0.5	0.1111
XGBoost	GHS01	safety_recall90	0.067481	18	0.9859	0.9743	0.9953	0.1507	0.0022	0.019	0.0011	1.0
XGBoost	GHS02	default_0.5	0.5	1063	0.9708	0.9654	0.9751	0.7552	0.5427	0.5638	0.3907	0.8881
XGBoost	GHS02	calibrated_F1	0.891765	1063	0.9708	0.9654	0.9751	0.7552	0.7071	0.694	0.7173	0.6971
XGBoost	GHS02	safety_recall90	0.535326	1063	0.9708	0.9654	0.9751	0.7552	0.5607	0.5776	0.4115	0.8796

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
XGBoost	GHS03	default_0.5	0.5	41	0.9267	0.8617	0.9807	0.5438	0.4355	0.4616	0.3253	0.6585
XGBoost	GHS03	calibrated_F1	0.978498	41	0.9267	0.8617	0.9807	0.5438	0.5797	0.5897	0.7143	0.4878
XGBoost	GHS03	safety_recall90	0.040324	41	0.9267	0.8617	0.9807	0.5438	0.2357	0.3228	0.1396	0.7561
XGBoost	GHS04	default_0.5	0.5	28	0.9951	0.9841	0.9999	0.8092	0.6667	0.6887	0.5319	0.8929
XGBoost	GHS04	calibrated_F1	0.969746	28	0.9951	0.9841	0.9999	0.8092	0.7368	0.7366	0.7241	0.75
XGBoost	GHS04	safety_recall90	0.730721	28	0.9951	0.9841	0.9999	0.8092	0.7188	0.7241	0.6389	0.8214
XGBoost	GHS05	default_0.5	0.5	4333	0.8924	0.8868	0.8979	0.7373	0.6304	0.545	0.5501	0.7381
XGBoost	GHS05	calibrated_F1	0.637968	4333	0.8924	0.8868	0.8979	0.7373	0.6654	0.5956	0.6837	0.648
XGBoost	GHS05	safety_recall90	0.258556	4333	0.8924	0.8868	0.8979	0.7373	0.5009	0.4122	0.3452	0.9128
XGBoost	GHS06	default_0.5	0.5	1520	0.8296	0.818	0.8402	0.3262	0.3353	0.3144	0.2273	0.6388
XGBoost	GHS06	calibrated_F1	0.633534	1520	0.8296	0.818	0.8402	0.3262	0.3612	0.3162	0.3352	0.3914
XGBoost	GHS06	safety_recall90	0.30901	1520	0.8296	0.818	0.8402	0.3262	0.2049	0.21	0.116	0.8803
XGBoost	GHS07	default_0.5	0.5	21967	0.8336	0.8261	0.8412	0.9765	0.8611	0.3401	0.9638	0.7781
XGBoost	GHS07	calibrated_F1	0.111349	21967	0.8336	0.8261	0.8412	0.9765	0.9502	0.202	0.9083	0.9962
XGBoost	GHS07	safety_recall90	0.295102	21967	0.8336	0.8261	0.8412	0.9765	0.9281	0.3268	0.9371	0.9193
XGBoost	GHS08	default_0.5	0.5	1360	0.9004	0.8927	0.9081	0.4518	0.3806	0.3833	0.2558	0.7434
XGBoost	GHS08	calibrated_F1	0.782329	1360	0.9004	0.8927	0.9081	0.4518	0.4668	0.4341	0.4406	0.4963
XGBoost	GHS08	safety_recall90	0.270562	1360	0.9004	0.8927	0.9081	0.4518	0.2765	0.3097	0.1638	0.8875
XGBoost	GHS09	default_0.5	0.5	1773	0.8969	0.8903	0.9033	0.4586	0.4334	0.418	0.3039	0.7552
XGBoost	GHS09	calibrated_F1	0.792764	1773	0.8969	0.8903	0.9033	0.4586	0.474	0.4307	0.4359	0.5195
XGBoost	GHS09	safety_recall90	0.295801	1773	0.8969	0.8903	0.9033	0.4586	0.371	0.3775	0.2367	0.8573
SVM	GHS01	default_0.5	0.5	18	0.9822	0.959	0.9984	0.4446	0.3509	0.3768	0.2564	0.5556
SVM	GHS01	calibrated_F1	0.922207	18	0.9822	0.959	0.9984	0.4446	0.3636	0.4713	1.0	0.2222
SVM	GHS01	safety_recall90	0.000997	18	0.9822	0.959	0.9984	0.4446	0.0029	0.0271	0.0015	1.0
SVM	GHS02	default_0.5	0.5	1063	0.934	0.9254	0.9418	0.5797	0.5254	0.5236	0.6801	0.428
SVM	GHS02	calibrated_F1	0.230508	1063	0.934	0.9254	0.9418	0.5797	0.5528	0.5398	0.4626	0.6867
SVM	GHS02	safety_recall90	0.024989	1063	0.934	0.9254	0.9418	0.5797	0.3226	0.3696	0.1976	0.8777
SVM	GHS03	default_0.5	0.5	41	0.9295	0.8628	0.9844	0.3229	0.2646	0.319	0.1689	0.6098
SVM	GHS03	calibrated_F1	0.808422	41	0.9295	0.8628	0.9844	0.3229	0.3621	0.3773	0.28	0.5122
SVM	GHS03	safety_recall90	0.02138	41	0.9295	0.8628	0.9844	0.3229	0.0826	0.1857	0.0435	0.8293
SVM	GHS04	default_0.5	0.5	28	0.9908	0.9725	0.9993	0.5309	0.3898	0.4572	0.2556	0.8214
SVM	GHS04	calibrated_F1	0.875264	28	0.9908	0.9725	0.9993	0.5309	0.5763	0.5765	0.5484	0.6071
SVM	GHS04	safety_recall90	0.316053	28	0.9908	0.9725	0.9993	0.5309	0.3268	0.4215	0.2	0.8929
SVM	GHS05	default_0.5	0.5	4333	0.8551	0.8486	0.8619	0.6395	0.5762	0.5158	0.7041	0.4877
SVM	GHS05	calibrated_F1	0.276596	4333	0.8551	0.8486	0.8619	0.6395	0.5915	0.4963	0.5427	0.6499
SVM	GHS05	safety_recall90	0.058622	4333	0.8551	0.8486	0.8619	0.6395	0.4542	0.3489	0.303	0.9065

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
SVM	GHS06	default_0.5	0.5	1520	0.7425	0.729	0.7555	0.1868	0.0117	0.0575	0.6429	0.0059
SVM	GHS06	calibrated_F1	0.134883	1520	0.7425	0.729	0.7555	0.1868	0.264	0.21	0.2069	0.3645
SVM	GHS06	safety_recall90	0.026945	1520	0.7425	0.729	0.7555	0.1868	0.1595	0.1322	0.0879	0.8618
SVM	GHS07	default_0.5	0.5	21967	0.7681	0.7582	0.7778	0.9647	0.9481	0.0867	0.904	0.9967
SVM	GHS07	calibrated_F1	0.316579	21967	0.7681	0.7582	0.7778	0.9647	0.9485	0.0281	0.9022	1.0
SVM	GHS07	safety_recall90	0.740947	21967	0.7681	0.7582	0.7778	0.9647	0.927	0.2452	0.9257	0.9283
SVM	GHS08	default_0.5	0.5	1360	0.8062	0.7948	0.8181	0.2579	0.0634	0.1208	0.5111	0.0338
SVM	GHS08	calibrated_F1	0.153138	1360	0.8062	0.7948	0.8181	0.2579	0.3392	0.2977	0.2956	0.3978
SVM	GHS08	safety_recall90	0.022405	1360	0.8062	0.7948	0.8181	0.2579	0.1728	0.1791	0.0958	0.8831
SVM	GHS09	default_0.5	0.5	1773	0.8319	0.8224	0.8414	0.3068	0.1831	0.1994	0.4484	0.1151
SVM	GHS09	calibrated_F1	0.292824	1773	0.8319	0.8224	0.8414	0.3068	0.3688	0.3173	0.3533	0.3858
SVM	GHS09	safety_recall90	0.035971	1773	0.8319	0.8224	0.8414	0.3068	0.2834	0.2755	0.1717	0.8111
RandomForest_NoClassWeight	GHS01	default_0.5	0.5	18	0.9891	0.9815	0.9956	0.3194	0.0	0.0	0.0	0.0
RandomForest_NoClassWeight	GHS01	calibrated_F1	0.089656	18	0.9891	0.9815	0.9956	0.3194	0.4167	0.4809	0.8333	0.2778
RandomForest_NoClassWeight	GHS01	safety_recall90	0.000488	18	0.9891	0.9815	0.9956	0.3194	0.0068	0.0519	0.0034	1.0
RandomForest_NoClassWeight	GHS02	default_0.5	0.5	1063	0.9638	0.9585	0.9686	0.7094	0.5356	0.5664	0.8571	0.3895
RandomForest_NoClassWeight	GHS02	calibrated_F1	0.280639	1063	0.9638	0.9585	0.9686	0.7094	0.6591	0.6433	0.6393	0.6802
RandomForest_NoClassWeight	GHS02	safety_recall90	0.08159	1063	0.9638	0.9585	0.9686	0.7094	0.46	0.4953	0.3092	0.8984
RandomForest_NoClassWeight	GHS03	default_0.5	0.5	41	0.9282	0.8657	0.9834	0.443	0.3265	0.4414	1.0	0.1951
RandomForest_NoClassWeight	GHS03	calibrated_F1	0.223356	41	0.9282	0.8657	0.9834	0.443	0.4262	0.4534	0.65	0.3171
RandomForest_NoClassWeight	GHS03	safety_recall90	0.007893	41	0.9282	0.8657	0.9834	0.443	0.0943	0.1939	0.0502	0.7805
RandomForest_NoClassWeight	GHS04	default_0.5	0.5	28	0.9782	0.9284	0.9996	0.6607	0.0	0.0	0.0	0.0
RandomForest_NoClassWeight	GHS04	calibrated_F1	0.124921	28	0.9782	0.9284	0.9996	0.6607	0.5783	0.611	0.4364	0.8571
RandomForest_NoClassWeight	GHS04	safety_recall90	0.100491	28	0.9782	0.9284	0.9996	0.6607	0.5275	0.5708	0.381	0.8571
RandomForest_NoClassWeight	GHS05	default_0.5	0.5	4333	0.8823	0.8767	0.888	0.7264	0.5372	0.5389	0.895	0.3838
RandomForest_NoClassWeight	GHS05	calibrated_F1	0.284526	4333	0.8823	0.8767	0.888	0.7264	0.6504	0.5766	0.6633	0.6379

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
RandomForest_NoClassWeight	GHS05	safety_recall90	0.108706	4333	0.8823	0.8767	0.888	0.7264	0.4733	0.3763	0.3194	0.9128
RandomForest_NoClassWeight	GHS06	default_0.5	0.5	1520	0.8293	0.8181	0.8396	0.3164	0.0208	0.0994	1.0	0.0105
RandomForest_NoClassWeight	GHS06	calibrated_F1	0.166874	1520	0.8293	0.8181	0.8396	0.3164	0.337	0.2966	0.3603	0.3164
RandomForest_NoClassWeight	GHS06	safety_recall90	0.038515	1520	0.8293	0.8181	0.8396	0.3164	0.1999	0.2043	0.1126	0.8882
RandomForest_NoClassWeight	GHS07	default_0.5	0.5	21967	0.8284	0.8204	0.8366	0.9753	0.95	0.1645	0.9051	0.9995
RandomForest_NoClassWeight	GHS07	calibrated_F1	0.645515	21967	0.8284	0.8204	0.8366	0.9753	0.9504	0.24	0.9117	0.9927
RandomForest_NoClassWeight	GHS07	safety_recall90	0.79188	21967	0.8284	0.8204	0.8366	0.9753	0.9283	0.3402	0.9391	0.9178
RandomForest_NoClassWeight	GHS08	default_0.5	0.5	1360	0.8917	0.8832	0.9006	0.4464	0.165	0.2715	0.8671	0.0912
RandomForest_NoClassWeight	GHS08	calibrated_F1	0.258257	1360	0.8917	0.8832	0.9006	0.4464	0.4314	0.4099	0.5175	0.3699
RandomForest_NoClassWeight	GHS08	safety_recall90	0.035427	1360	0.8917	0.8832	0.9006	0.4464	0.2401	0.2721	0.1385	0.9022
RandomForest_NoClassWeight	GHS09	default_0.5	0.5	1773	0.8794	0.8718	0.8873	0.4496	0.1399	0.2249	0.7405	0.0773
RandomForest_NoClassWeight	GHS09	calibrated_F1	0.231742	1773	0.8794	0.8718	0.8873	0.4496	0.4638	0.4196	0.4273	0.5071
RandomForest_NoClassWeight	GHS09	safety_recall90	0.081373	1773	0.8794	0.8718	0.8873	0.4496	0.3642	0.364	0.2338	0.824

This table has 22 columns in full; 13 are shown here and 9 are omitted so the table stays legible. The complete table, with every column, is sheet 'S3_full_performance' of the accompanying Excel workbook.

108 rows. Source sheet: S3_full_performance.

Table S4

The twenty most influential SHAP features per hazard class.

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP
GHS01_Explosive	1	MACCS_70	0.55366	-0.463924
GHS01_Explosive	2	MACCS_124	0.490891	-0.323515
GHS01_Explosive	3	TPSA	0.275038	-0.269869
GHS01_Explosive	4	MACCS_142	0.166226	-0.160253
GHS01_Explosive	5	MACCS_102	0.140569	-0.039449
GHS01_Explosive	6	Chi2n	0.075281	-0.061007
GHS01_Explosive	7	Kappa3	0.070183	-0.062226
GHS01_Explosive	8	MACCS_49	0.067073	-0.058121
GHS01_Explosive	9	MACCS_148	0.060744	-0.005539
GHS01_Explosive	10	MACCS_83	0.055498	-0.024283
GHS01_Explosive	11	MACCS_90	0.045773	-0.044074
GHS01_Explosive	12	MACCS_156	0.035559	-0.020667
GHS01_Explosive	13	Morgan_514	0.035028	-0.034336
GHS01_Explosive	14	MACCS_154	0.034543	-0.025669
GHS01_Explosive	15	Morgan_128	0.030807	-0.01278
GHS01_Explosive	16	MACCS_117	0.03029	-0.013108
GHS01_Explosive	17	MACCS_111	0.02967	-0.020643
GHS01_Explosive	18	MACCS_121	0.029436	-0.021602
GHS01_Explosive	19	MACCS_161	0.028044	-0.027391
GHS01_Explosive	20	NumHAcceptors	0.027608	-0.025232
GHS02_Flammable	1	BertzCT	0.649036	-0.375776
GHS02_Flammable	2	TPSA	0.540827	-0.349749
GHS02_Flammable	3	LabuteASA	0.439609	-0.25814
GHS02_Flammable	4	NumHDonors	0.233505	-0.07345
GHS02_Flammable	5	NHOHCount	0.181492	-0.073873
GHS02_Flammable	6	MolWt	0.172027	-0.122351
GHS02_Flammable	7	MACCS_131	0.169805	-0.109887
GHS02_Flammable	8	ExactMolWt	0.161357	-0.122906
GHS02_Flammable	9	Morgan_650	0.152775	-0.052726
GHS02_Flammable	10	FractionCSP3	0.152108	-0.069067
GHS02_Flammable	11	Morgan_356	0.138924	-0.059179
GHS02_Flammable	12	MACCS_103	0.120147	-0.028014
GHS02_Flammable	13	Chi4n	0.117017	-0.087387

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP
GHS02_Flammable	14	NumHeteroatoms	0.090413	-0.023964
GHS02_Flammable	15	MACCS_154	0.087076	-0.030397
GHS02_Flammable	16	MACCS_88	0.087028	0.000504
GHS02_Flammable	17	RingCount	0.084934	-0.024075
GHS02_Flammable	18	MACCS_124	0.083471	-0.022454
GHS02_Flammable	19	MACCS_149	0.081658	-0.008876
GHS02_Flammable	20	Chi3n	0.080891	-0.054113
GHS03_Oxidising	1	Chi3n	0.952689	-0.900104
GHS03_Oxidising	2	MACCS_102	0.77605	-0.539722
GHS03_Oxidising	3	TPSA	0.611993	-0.58199
GHS03_Oxidising	4	MACCS_124	0.455176	-0.396334
GHS03_Oxidising	5	Kappa3	0.446737	-0.442095
GHS03_Oxidising	6	FractionCSP3	0.405366	-0.404242
GHS03_Oxidising	7	MACCS_31	0.375778	-0.310106
GHS03_Oxidising	8	MolLogP	0.373483	-0.373384
GHS03_Oxidising	9	BalabanJ	0.365839	-0.365465
GHS03_Oxidising	10	Chi4n	0.350112	-0.331174
GHS03_Oxidising	11	Kappa2	0.296654	-0.294025
GHS03_Oxidising	12	BertzCT	0.281685	-0.281145
GHS03_Oxidising	13	MACCS_49	0.2722	-0.182807
GHS03_Oxidising	14	MACCS_146	0.261383	-0.1793
GHS03_Oxidising	15	MACCS_88	0.234421	-0.084734
GHS03_Oxidising	16	LabuteASA	0.229907	-0.229907
GHS03_Oxidising	17	Kappa1	0.227362	-0.226283
GHS03_Oxidising	18	Morgan_378	0.221343	-0.145816
GHS03_Oxidising	19	MACCS_130	0.214334	-0.14282
GHS03_Oxidising	20	MolWt	0.203915	-0.203915
GHS04_CompressedGas	1	Kappa2	1.79048	-1.78889
GHS04_CompressedGas	2	Chi4n	1.539784	-1.39107
GHS04_CompressedGas	3	LabuteASA	1.355818	-1.354701
GHS04_CompressedGas	4	MolWt	0.767101	-0.766134
GHS04_CompressedGas	5	ExactMolWt	0.74815	-0.746992
GHS04_CompressedGas	6	Chi2n	0.684677	-0.682578
GHS04_CompressedGas	7	TPSA	0.661865	-0.649623
GHS04_CompressedGas	8	BertzCT	0.62329	-0.621016
GHS04_CompressedGas	9	Chi3n	0.53796	-0.53156

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP
GHS04_CompressedGas	10	MolLogP	0.520711	-0.520711
GHS04_CompressedGas	11	NumRotatableBonds	0.467301	-0.438768
GHS04_CompressedGas	12	Kappa3	0.448049	-0.438488
GHS04_CompressedGas	13	NumHDonors	0.417581	-0.399102
GHS04_CompressedGas	14	NumHeteroatoms	0.382897	-0.303319
GHS04_CompressedGas	15	BalabanJ	0.375097	-0.371112
GHS04_CompressedGas	16	Morgan_356	0.294656	-0.287391
GHS04_CompressedGas	17	NumHAcceptors	0.292755	-0.272068
GHS04_CompressedGas	18	Morgan_33	0.292084	-0.231596
GHS04_CompressedGas	19	Kappa1	0.291974	0.100052
GHS04_CompressedGas	20	NumAromaticRings	0.239348	-0.236859
GHS05_Corrosive	1	BertzCT	0.345213	-0.102352
GHS05_Corrosive	2	Morgan_514	0.235701	-0.120401
GHS05_Corrosive	3	MACCS_31	0.153432	-0.027762
GHS05_Corrosive	4	MACCS_154	0.12328	0.000308
GHS05_Corrosive	5	MACCS_153	0.113646	-0.037132
GHS05_Corrosive	6	MACCS_117	0.096806	-0.00408
GHS05_Corrosive	7	RingCount	0.092477	-0.023207
GHS05_Corrosive	8	MACCS_139	0.091429	-0.010479
GHS05_Corrosive	9	BalabanJ	0.086371	-0.030852
GHS05_Corrosive	10	TPSA	0.084789	-0.007402
GHS05_Corrosive	11	MACCS_100	0.079445	-0.004686
GHS05_Corrosive	12	Morgan_659	0.070868	-0.034967
GHS05_Corrosive	13	MACCS_96	0.068252	-0.020786
GHS05_Corrosive	14	MACCS_151	0.067597	0.004427
GHS05_Corrosive	15	FractionCSP3	0.067205	-0.009668
GHS05_Corrosive	16	MACCS_158	0.066705	-0.001137
GHS05_Corrosive	17	Morgan_85	0.065138	-0.011784
GHS05_Corrosive	18	MACCS_84	0.062641	-0.008013
GHS05_Corrosive	19	Morgan_251	0.061808	-0.006315
GHS05_Corrosive	20	Morgan_356	0.06107	0.001274
GHS06_AcuteToxicity	1	ExactMolWt	0.171272	-0.080114
GHS06_AcuteToxicity	2	MACCS_41	0.090883	-0.039571
GHS06_AcuteToxicity	3	MACCS_164	0.077375	-0.012709
GHS06_AcuteToxicity	4	Chi3n	0.076763	-0.020702
GHS06_AcuteToxicity	5	Chi2n	0.067517	-0.018624

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP
GHS06_AcuteToxicity	6	BalabanJ	0.052652	-0.023093
GHS06_AcuteToxicity	7	MACCS_119	0.051779	-0.008637
GHS06_AcuteToxicity	8	MACCS_123	0.050552	-0.015656
GHS06_AcuteToxicity	9	MACCS_137	0.047352	-0.030456
GHS06_AcuteToxicity	10	Chi1	0.044922	-0.009751
GHS06_AcuteToxicity	11	Morgan_926	0.040315	-0.010392
GHS06_AcuteToxicity	12	MACCS_157	0.035443	-0.008175
GHS06_AcuteToxicity	13	MolWt	0.032215	-0.020934
GHS06_AcuteToxicity	14	Chi4n	0.031492	-0.016013
GHS06_AcuteToxicity	15	BertzCT	0.029485	-0.022315
GHS06_AcuteToxicity	16	Chi0	0.027305	-0.001726
GHS06_AcuteToxicity	17	TPSA	0.026778	-0.009967
GHS06_AcuteToxicity	18	MACCS_120	0.026268	-0.010285
GHS06_AcuteToxicity	19	NHOHCount	0.025909	-0.007294
GHS06_AcuteToxicity	20	FractionCSP3	0.02488	-0.012069
GHS07_Irritant	1	Chi1	0.164398	0.061072
GHS07_Irritant	2	MACCS_137	0.153679	0.058604
GHS07_Irritant	3	LabuteASA	0.13757	0.047797
GHS07_Irritant	4	BertzCT	0.11732	0.038474
GHS07_Irritant	5	MolLogP	0.115468	0.020777
GHS07_Irritant	6	RingCount	0.105317	0.034953
GHS07_Irritant	7	Morgan_514	0.098046	0.037188
GHS07_Irritant	8	MACCS_121	0.097865	0.018133
GHS07_Irritant	9	NumHDonors	0.090397	0.02289
GHS07_Irritant	10	MACCS_65	0.077758	0.019894
GHS07_Irritant	11	MACCS_31	0.060578	0.021937
GHS07_Irritant	12	MACCS_154	0.058368	0.013593
GHS07_Irritant	13	MolWt	0.056935	0.027487
GHS07_Irritant	14	MACCS_139	0.052449	0.002712
GHS07_Irritant	15	MACCS_123	0.048644	0.017686
GHS07_Irritant	16	NumHeteroatoms	0.046305	0.018333
GHS07_Irritant	17	ExactMolWt	0.044717	0.023948
GHS07_Irritant	18	Chi4n	0.043983	0.021953
GHS07_Irritant	19	Chi3n	0.042163	0.02033
GHS07_Irritant	20	TPSA	0.042063	0.015136
GHS08_HealthHazard	1	Chi4n	0.175769	-0.100279

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP
GHS08_HealthHazard	2	TPSA	0.140887	-0.071943
GHS08_HealthHazard	3	NumHDonors	0.121378	-0.03439
GHS08_HealthHazard	4	MACCS_121	0.118889	-0.027694
GHS08_HealthHazard	5	MACCS_119	0.115361	-0.065893
GHS08_HealthHazard	6	Chi1	0.106844	-0.037051
GHS08_HealthHazard	7	Chi2n	0.09944	-0.053133
GHS08_HealthHazard	8	Morgan_893	0.099022	-0.020896
GHS08_HealthHazard	9	Morgan_725	0.094818	-0.035108
GHS08_HealthHazard	10	Chi3n	0.083487	-0.056616
GHS08_HealthHazard	11	MACCS_44	0.079207	-0.038349
GHS08_HealthHazard	12	Chi0	0.076798	-0.037701
GHS08_HealthHazard	13	NumHeteroatoms	0.0735	-0.041089
GHS08_HealthHazard	14	RingCount	0.071652	-0.02386
GHS08_HealthHazard	15	Kappa1	0.071568	-0.029936
GHS08_HealthHazard	16	MACCS_134	0.069779	-0.003579
GHS08_HealthHazard	17	ExactMolWt	0.064652	-0.031146
GHS08_HealthHazard	18	LabuteASA	0.061388	-0.019984
GHS08_HealthHazard	19	MolWt	0.060529	-0.036814
GHS08_HealthHazard	20	FractionCSP3	0.058271	-0.030057
GHS09_Environmental	1	MolLogP	0.378431	-0.133164
GHS09_Environmental	2	MACCS_137	0.358769	-0.145773
GHS09_Environmental	3	Chi1	0.30195	-0.145785
GHS09_Environmental	4	LabuteASA	0.25488	-0.145089
GHS09_Environmental	5	MACCS_65	0.129309	-0.055423
GHS09_Environmental	6	NumAromaticRings	0.128962	-0.031911
GHS09_Environmental	7	Morgan_514	0.12152	-0.055874
GHS09_Environmental	8	MACCS_44	0.112232	-0.040265
GHS09_Environmental	9	MACCS_18	0.103174	-0.072472
GHS09_Environmental	10	FractionCSP3	0.102216	-0.039803
GHS09_Environmental	11	Kappa2	0.098929	-0.054517
GHS09_Environmental	12	MACCS_46	0.085896	-0.015112
GHS09_Environmental	13	Morgan_378	0.078924	-0.00221
GHS09_Environmental	14	TPSA	0.071746	-0.017135
GHS09_Environmental	15	Morgan_389	0.07161	0.000231
GHS09_Environmental	16	MACCS_134	0.068983	-0.004196
GHS09_Environmental	17	MACCS_145	0.066914	-0.033925

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP
GHS09_Environmental	18	NumHeteroatoms	0.0636	-0.018185
GHS09_Environmental	19	Chi0	0.058698	-0.017893
GHS09_Environmental	20	Chi4n	0.058114	-0.02602

This table has 9 columns in full; 5 are shown here and 4 are omitted so the table stays legible. The complete table, with every column, is sheet 'S4_SHAP_top20' of the accompanying Excel workbook.

180 rows. Source sheet: S4_SHAP_top20.

Table S4b

Chemical interpretation of the five most influential features per hazard class.

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS01	1	MACCS_70	0.55366	positive - a HIGHER value of th...	MACCS substructure key 70: matc...	Explosivity comes from function...	not among the classically expec...
GHS01	2	MACCS_124	0.490891	positive - a HIGHER value of th...	MACCS substructure key 124: mat...	Explosivity comes from function...	not among the classically expec...
GHS01	3	TPSA	0.275038	positive - a HIGHER value of th...	topological polar surface area ...	Explosivity comes from function...	not among the classically expec...
GHS01	4	MACCS_142	0.166226	positive - a HIGHER value of th...	MACCS substructure key 142: mat...	Explosivity comes from function...	not among the classically expec...
GHS01	5	MACCS_102	0.140569	positive - a HIGHER value of th...	MACCS substructure key 102: mat...	Explosivity comes from function...	not among the classically expec...
GHS02	1	BertzCT	0.649036	negative - a HIGHER value of th...	structural complexity	Flammability tracks volatility ...	not among the classically expec...
GHS02	2	TPSA	0.540827	negative - a HIGHER value of th...	topological polar surface area ...	Flammability tracks volatility ...	yes - this is a descriptor a ch...
GHS02	3	LabuteASA	0.439609	negative - a HIGHER value of th...	approximate molecular surface area	Flammability tracks volatility ...	not among the classically expec...
GHS02	4	NumHDonors	0.233505	negative - a HIGHER value of th...	number of hydrogen-bond donors ...	Flammability tracks volatility ...	not among the classically expec...
GHS02	5	NHOHCount	0.181492	negative - a HIGHER value of th...	number of NH and OH groups	Flammability tracks volatility ...	not among the classically expec...
GHS03	1	Chi3n	0.952689	negative - a HIGHER value of th...	third-order connectivity index	Oxidisers carry oxygen in a hig...	not among the classically expec...
GHS03	2	MACCS_102	0.77605	positive - a HIGHER value of th...	MACCS substructure key 102: mat...	Oxidisers carry oxygen in a hig...	not among the classically expec...
GHS03	3	TPSA	0.611993	positive - a HIGHER value of th...	topological polar surface area ...	Oxidisers carry oxygen in a hig...	not among the classically expec...
GHS03	4	MACCS_124	0.455176	positive - a HIGHER value of th...	MACCS substructure key 124: mat...	Oxidisers carry oxygen in a hig...	not among the classically expec...
GHS03	5	Kappa3	0.446737	non-monotonic - the descriptor ...	third-order shape index - centr...	Oxidisers carry oxygen in a hig...	not among the classically expec...
GHS04	1	Kappa2	1.79048	negative - a HIGHER value of th...	second-order shape index - how ...	The compressed-gas pictogram is...	not among the classically expec...
GHS04	2	Chi4n	1.539784	negative - a HIGHER value of th...	fourth-order connectivity index	The compressed-gas pictogram is...	not among the classically expec...
GHS04	3	LabuteASA	1.355818	negative - a HIGHER value of th...	approximate molecular surface area	The compressed-gas pictogram is...	yes - this is a descriptor a ch...
GHS04	4	MolWt	0.767101	negative - a HIGHER value of th...	average molecular mass	The compressed-gas pictogram is...	yes - this is a descriptor a ch...
GHS04	5	ExactMolWt	0.74815	negative - a HIGHER value of th...	monoisotopic molecular mass	The compressed-gas pictogram is...	yes - this is a descriptor a ch...

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS05	1	BertzCT	0.345213	negative - a HIGHER value of th...	structural complexity	Corrosivity is driven by extrem...	not among the classically expec...
GHS05	2	Morgan_514	0.235701	negative - a HIGHER value of th...	ECFP4 circular substructure bit...	Corrosivity is driven by extrem...	not among the classically expec...
GHS05	3	MACCS_31	0.153432	positive - a HIGHER value of th...	MACCS substructure key 31: matc...	Corrosivity is driven by extrem...	not among the classically expec...
GHS05	4	MACCS_154	0.12328	negative - a HIGHER value of th...	MACCS substructure key 154: mat...	Corrosivity is driven by extrem...	not among the classically expec...
GHS05	5	MACCS_153	0.113646	positive - a HIGHER value of th...	MACCS substructure key 153: mat...	Corrosivity is driven by extrem...	not among the classically expec...
GHS06	1	ExactMolWt	0.171272	positive - a HIGHER value of th...	monoisotopic molecular mass	Acute toxicity requires a molec...	not among the classically expec...
GHS06	2	MACCS_41	0.090883	positive - a HIGHER value of th...	MACCS substructure key 41: matc...	Acute toxicity requires a molec...	not among the classically expec...
GHS06	3	MACCS_164	0.077375	negative - a HIGHER value of th...	MACCS substructure key 164: mat...	Acute toxicity requires a molec...	not among the classically expec...
GHS06	4	Chi3n	0.076763	negative - a HIGHER value of th...	third-order connectivity index	Acute toxicity requires a molec...	not among the classically expec...
GHS06	5	Chi2n	0.067517	negative - a HIGHER value of th...	second-order connectivity index	Acute toxicity requires a molec...	not among the classically expec...
GHS07	1	Chi1	0.164398	negative - a HIGHER value of th...	first-order connectivity index ...	The exclamation-mark pictogram ...	not among the classically expec...
GHS07	2	MACCS_137	0.153679	positive - a HIGHER value of th...	MACCS substructure key 137: mat...	The exclamation-mark pictogram ...	not among the classically expec...
GHS07	3	LabuteASA	0.13757	negative - a HIGHER value of th...	approximate molecular surface area	The exclamation-mark pictogram ...	not among the classically expec...
GHS07	4	BertzCT	0.11732	negative - a HIGHER value of th...	structural complexity	The exclamation-mark pictogram ...	not among the classically expec...
GHS07	5	MolLogP	0.115468	negative - a HIGHER value of th...	lipophilicity - how strongly th...	The exclamation-mark pictogram ...	yes - this is a descriptor a ch...
GHS08	1	Chi4n	0.175769	positive - a HIGHER value of th...	fourth-order connectivity index	The health-hazard pictogram fla...	not among the classically expec...
GHS08	2	TPSA	0.140887	positive - a HIGHER value of th...	topological polar surface area ...	The health-hazard pictogram fla...	not among the classically expec...
GHS08	3	NumHDonors	0.121378	negative - a HIGHER value of th...	number of hydrogen-bond donors ...	The health-hazard pictogram fla...	not among the classically expec...
GHS08	4	MACCS_121	0.118889	negative - a HIGHER value of th...	MACCS substructure key 121: mat...	The health-hazard pictogram fla...	not among the classically expec...
GHS08	5	MACCS_119	0.115361	positive - a HIGHER value of th...	MACCS substructure key 119: mat...	The health-hazard pictogram fla...	not among the classically expec...
GHS09	1	MolLogP	0.378431	positive - a HIGHER value of th...	lipophilicity - how strongly th...	The environmental pictogram is ...	yes - this is a descriptor a ch...
GHS09	2	MACCS_137	0.358769	negative - a HIGHER value of th...	MACCS substructure key 137: mat...	The environmental pictogram is ...	not among the classically expec...

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS09	3	Chi1	0.30195	positive - a HIGHER value of th...	first-order connectivity index ...	The environmental pictogram is ...	not among the classically expec...
GHS09	4	LabuteASA	0.25488	positive - a HIGHER value of th...	approximate molecular surface area	The environmental pictogram is ...	not among the classically expec...
GHS09	5	MACCS_65	0.129309	negative - a HIGHER value of th...	MACCS substructure key 65: matc...	The environmental pictogram is ...	not among the classically expec...

This table has 12 columns in full; 8 are shown here and 4 are omitted so the table stays legible. The complete table, with every column, is sheet 'S4b_SHAP_interpretation' of the accompanying Excel workbook.

45 rows. Source sheet: S4b_SHAP_interpretation.

Table S5

Malaysian industrial validation, per hazard class.

GHS Column	Pictogram Code	Meaning	N True Positive In Set	Accuracy	Precision	Recall	F1	MCC
GHS01_Explosive	GHS01	Explosive (exploding bomb)	0	1.0	0.0	0.0	0.0	
GHS02_Flammable	GHS02	Flammable (flame)	28	0.8837	0.871	0.9643	0.9153	0.7413
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	2	1.0	1.0	1.0	1.0	1.0
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	10	0.9535	0.8333	1.0	0.9091	0.8848
GHS05_Corrosive	GHS05	Corrosive (corrosion)	14	0.8605	0.8333	0.7143	0.7692	0.6742
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and cross...	16	0.7907	0.6522	0.9375	0.7692	0.6214
GHS07_Irritant	GHS07	Irritant / harmful (exclamation...	37	0.8605	0.9429	0.8919	0.9167	0.4974
GHS08_HealthHazard	GHS08	Serious health hazard (health h...	36	0.7674	0.8611	0.8611	0.8611	0.1468
GHS09_Environmental	GHS09	Environmental hazard (environment)	25	0.4884	0.8	0.16	0.2667	0.1607

9 rows. Source sheet: S5_malaysia_per_class.

Table S5b

Malaysian industrial validation, per sector.

Sector	N Compounds	Label Accuracy	Hazard Recall	N Actual Hazard Labels	N Hazards Correctly Flagged
Johor 2019 Emergency	12	0.9074	0.8364	55	46
Palm Oil Processing	7	0.9048	0.9	20	18
Petrochemicals	9	0.8765	0.8387	31	26
Rubber Processing	7	0.7619	0.7083	24	17
Semiconductor (Penang)	8	0.7361	0.6579	38	25

5 rows. Source sheet: S5b_malaysia_per_sector.

Table S5c

The chemicals implicated in the 2019 Sungai Kim Kim incident at Pasir Gudang, Johor, with predicted and reference hazard profiles.

Query Name	PubChem Name	CID	MolecularFormula	Sector	Incident Role	Substitution Note
acrylonitrile	Acrylonitrile	7855	C3H3N	Johor 2019 Emergency	principal agent identified by t...	
acrolein	Acrolein	7847	C3H4O	Johor 2019 Emergency	principal agent; a severe respi...	
benzene	Benzene	241	C6H6	Johor 2019 Emergency	volatile aromatic detected in a...	
toluene	Toluene	1140	C7H8	Johor 2019 Emergency	volatile aromatic detected in a...	
p-xylene	P-Xylene	7809	C8H10	Johor 2019 Emergency	volatile aromatic detected in a...	'xylene' is an isomer mixture r...
ethylbenzene	Ethylbenzene	7500	C8H10	Johor 2019 Emergency	volatile aromatic detected in a...	
hydrogen sulfide	Hydrogen Sulfide	402	H2S	Johor 2019 Emergency	released from the anaerobic riv...	
methane	Methane	297	CH4	Johor 2019 Emergency	released from the anaerobic riv...	
methyl mercaptan	Methanethiol	878	CH4S	Johor 2019 Emergency	odorous sulfur compound reporte...	
d-limonene	(+)-Limonene	440917	C10H16	Johor 2019 Emergency	solvent component reported in t...	
hydrogen chloride	Hydrochloric Acid	313	ClH	Johor 2019 Emergency	acid gas reported in the waste ...	
styrene	Styrene	7501	C8H8	Johor 2019 Emergency	monomer reported among the solv...	

This table has 38 columns in full; 7 are shown here and 31 are omitted so the table stays legible. The complete table, with every column, is sheet 'S5c_johor_2019' of the accompanying Excel workbook.

12 rows. Source sheet: S5c_johor_2019.

File S1

Complete software environment. Every version is pinned, and a single random seed of 42 is applied throughout, so the analysis reproduces exactly.

```
# ==========================================================================
# STEP1_environment_requirements.txt
# GHS Chemical Hazard Classification project - Sareer Ahmad
# Generated: 2026-08-06T06:22:08
# Python   : 3.11.15 (D:\GHS_Project\.venv\Scripts\python.exe)
# Platform : Windows-10-10.0.19045-SP0
# Random seed used project-wide: 42
#
# INSTALLATION NOTE (documented deviation, see Rule 2):
#   The official python.org Windows installer (python-3.11.9-amd64.exe)
#   failed twice with WiX bootstrapper exit code 0x3 on this machine
#   (no administrator elevation available for the chained MSI packages).
#   FALLBACK APPLIED: the 'uv' package manager (Astral) was used to
#   install a standalone CPython 3.11.15 build, which requires no
#   Windows installer at all. All packages were then installed with
#   'uv pip install', which is a drop-in replacement for pip.
#   conda was not available on this machine, so the conda fallback
#   named in the proposal could not be used.
# ==========================================================================

# ==========================================================================
# PINNED DEPENDENCIES - DO NOT UPGRADE WITHOUT READING THIS
# ==========================================================================
#
# starlette==0.52.1
#   WHY: Streamlit 1.61.0 declares 'starlette<2,>=0.46.0', but that range is
#        too permissive. Starlette 1.4.0 added a required keyword-only
#        argument 'thread_minimum_size' to GZipResponder.__init__, which
#        Streamlit's own gzip middleware subclasses without passing. With
#        Starlette >= 1.0 installed, every HTTP response from the web
#        application fails with HTTP 500 and the page never loads. Installing
#        with an unconstrained resolver picks the newest Starlette and
#        reproduces the fault, so the pin is required.
#   SYMPTOM IF UNPINNED: TypeError: GZipResponder.__init__() missing 1 required keyword-only argument: 'thread_minimum_size'
#   VERIFIED WORKING: streamlit 1.61.0 + starlette 0.52.1
#
starlette<1.0
# ==========================================================================

# --- Directly required by the research proposal ---
requests==2.34.2
pandas==3.0.5
numpy==2.4.6
rdkit==2026.03.5
scikit-learn==1.9.0
xgboost==3.2.0
imbalanced-learn==0.14.2
shap==0.51.0
matplotlib==3.11.1
seaborn==0.13.2
```

```
scipy==1.17.1
joblib==1.5.3
streamlit==1.61.0
reportlab==5.0.0
tqdm==4.70.0
pubchempy==1.0.5
lightgbm==4.7.0
iterative-stratification==0.1.9
openpyxl==3.1.5
starlette==0.52.1


# --- Complete frozen environment (pip freeze) ---
```